

TURTLES AND TORTOISES

With their bony shell, stout limbs, and toothless, beak-like mouth, turtles and tortoises have changed little from species that lived 200 million years ago.

The order includes sea turtles, freshwater turtles, and terrestrial tortoises. Some extinct turtles were gigantic, but most modern ones are more modestly sized. The exceptions are the large sea turtles and a few terrestrial tortoises inhabiting isolated islands.

PROTECTION AND LOCOMOTION

A turtle's shell is formed of numerous fused bones, overlaid by scutes. The domed upperpart is known as the carapace and the lower part, the plastron. Scutes are horny plates and new ones form beneath the old ones each year. The degree of protection provided by the shell varies between species and is much reduced in some aquatic turtles. Not all species can retract the head into the shell – some tuck their head along one shoulder under the edge of the shell.

Tortoises are slow animals, but some sea turtles can reach speeds of 30 kph (19 mph) when swimming, as a result of the modification of their

forelimbs into flippers. Although they need to breathe air, many turtles are tolerant of low oxygen levels and can remain submerged for hours. Turtles have a slow metabolism and are generally long lived.

Some species are carnivores, others herbivores, but most turtles are omnivorous. Animal prey is either slow moving or ambushed as the turtle lies hidden. Food is broken up by the “beak”, a sharp, keratinous jaw covering.

NESTING AND BREEDING

Turtles do not defend territories, but they can have extensive home ranges and may develop social hierarchies. Otherwise turtles often congregate on river banks or lakesides to bask or nest.

Both in water and on land, males of some species engage in elaborate courtship of females before copulation. Fertilization is internal and the females lay shelled eggs, like those of other reptiles and birds. These are spherical or elongated, with rigid or flexible shells. Clutches are laid in nests dug up by the female in terrestrial sites. Nearly all marine species come to land only to nest. In many species, but not all, the gender of the hatchlings depends on incubation temperature.

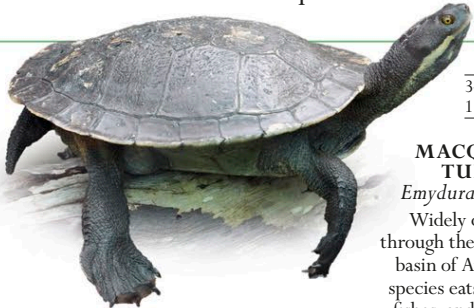
PHYLUM	CHORDATA
CLASS	REPTILIA
ORDER	TESTUDINES
FAMILIES	14
SPECIES	353



A green sea turtle hatchling enters the sea. Many of its breeding beaches are protected, but it remains endangered.

AUSTRO-AMERICAN SIDE-NECKED TURTLES

From South America and Australasia, the Chelidae include carnivorous and omnivorous species. Their characteristically long neck cannot be retracted, so it is turned sideways under the edge of the shell. They lay elongated eggs with leathery shells.



34 cm
13½ in

MACQUARIE TURTLE

Emydura macquarii
Widely distributed through the Murray River basin of Australia, this species eats amphibians, fishes, and algae. Males are smaller than females.



75 cm
30 in

REIMANN'S SNAKE-NECKED TURTLE

Chelodina reimanni

This turtle from New Guinea eats crustaceans and molluscs. When threatened, it tucks its large head under the side of its carapace.

scalloped carapace with keel (central ridge)

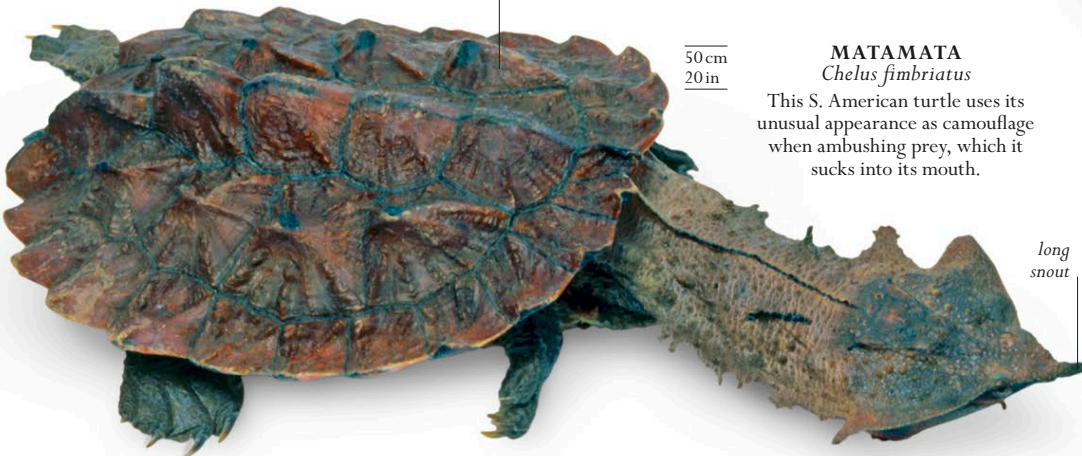


25 cm
10 in

COMMON SNAKE-NECKED TURTLE

Chelodina longicollis

A shy freshwater turtle of Australia, this species has a long neck, which enables it to raise its head out of water and capture prey.



50 cm
20 in

MATAMATA

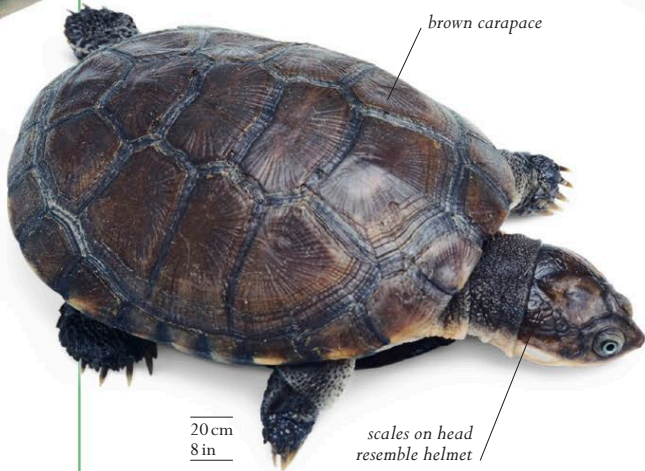
Chelus fimbriatus

This S. American turtle uses its unusual appearance as camouflage when ambushing prey, which it sucks into its mouth.

long snout

AFRICAN SIDE-NECKED TURTLES

Most of the Pelomedusidae are carnivorous and occupy freshwater habitats. When threatened, they can hide their head and neck beneath the edge of the shell. Found throughout Africa and Madagascar, they survive dry conditions by burying themselves in mud.



20 cm
8 in

AFRICAN HELMETED TURTLE

Pelomedusa subrufa

Widespread in sub-Saharan Africa, this carnivorous turtle is highly sociable and often hunts in packs to bring down large prey.